

02YMECH - Solution 4

Assigned for the week of Oct 13, 2025

Solutions

1. Time to first maximum displacement (undamped oscillator).

The motion is

$$x(t) = A \cos(\omega_0 t + \phi),$$

where A , ω_0 , and ϕ are constants. The velocity is

$$v(t) = \dot{x}(t) = -A\omega_0 \sin(\omega_0 t + \phi).$$

Maximum displacement in simple harmonic motion occurs when $|x(t)| = A$, i.e. when the cosine is ± 1 . At those instants, the velocity is zero.

We are told the initial position and velocity:

$$x(0) = x_0 = A \cos \phi, \quad v(0) = v_0 = -A\omega_0 \sin \phi.$$

From these,

$$\cos \phi = \frac{x_0}{A}, \quad \sin \phi = -\frac{v_0}{A\omega_0}.$$

We want the first time $t > 0$ such that $x(t)$ reaches an *extreme* (turning point), i.e. when $v(t) = 0$ and x is a maximum in magnitude.

The condition $v(t) = 0$ gives

$$\sin(\omega_0 t + \phi) = 0 \implies \omega_0 t + \phi = n\pi, \quad n \in \mathbb{Z}.$$

Thus the candidate times are

$$t_n = \frac{n\pi - \phi}{\omega_0}.$$

At those instants,

$$x(t_n) = A \cos(n\pi) = A(-1)^n.$$

So for even n ($n = 0, 2, 4, \dots$), $x = +A$; for odd n ($n = 1, 3, 5, \dots$), $x = -A$. Both $\pm A$ are extrema of equal magnitude.

We want the *first* $t > 0$. The smallest $t_n > 0$ is obtained by choosing the smallest integer n such that

$$t_n = \frac{n\pi - \phi}{\omega_0} > 0.$$

This gives the rule:

$$t_{\text{first}} = \min_{n \in \mathbb{Z}} \left\{ \frac{n\pi - \phi}{\omega_0} \mid \frac{n\pi - \phi}{\omega_0} > 0 \right\}.$$

In words: find the next zero of the sine after $t = 0$. Equivalently, if $\phi \in (-\pi, \pi]$,

$$t_{\text{first}} = \begin{cases} \frac{-\phi}{\omega_0}, & \text{if } \phi < 0, \\ \frac{\pi - \phi}{\omega_0}, & \text{if } \phi \geq 0. \end{cases}$$

We can also express ϕ in terms of x_0, v_0 using

$$\phi = \arctan 2\left(-\frac{v_0}{A\omega_0}, \frac{x_0}{A}\right),$$

where $\arctan 2(y, x)$ is the correct quadrant angle with $\cos \phi = x_0/A$ and $\sin \phi = -v_0/(A\omega_0)$.

Summary: the first time the oscillator reaches an endpoint ($|x| = A$) after $t = 0$ is given by the smallest positive member of $t_n = (n\pi - \phi)/\omega_0$.

Physically: the oscillator reaches maximum displacement whenever its velocity first becomes zero.

2. Average kinetic and potential energy over one cycle.

For a harmonic oscillator of mass m and angular frequency ω_0 , we can write

$$x(t) = A \cos(\omega_0 t + \phi).$$

Then

$$v(t) = \dot{x}(t) = -A\omega_0 \sin(\omega_0 t + \phi).$$

The kinetic energy is

$$K(t) = \frac{1}{2}mv^2(t) = \frac{1}{2}mA^2\omega_0^2 \sin^2(\omega_0 t + \phi).$$

The potential energy (spring potential) is

$$U(t) = \frac{1}{2}kx^2(t) = \frac{1}{2}kA^2 \cos^2(\omega_0 t + \phi),$$

with $k = m\omega_0^2$ for a simple harmonic oscillator. Thus

$$U(t) = \frac{1}{2}m\omega_0^2 A^2 \cos^2(\omega_0 t + \phi).$$

The total mechanical energy is constant:

$$E = K(t) + U(t) = \frac{1}{2}m\omega_0^2 A^2.$$

We now compute time averages over one full period $T = \frac{2\pi}{\omega_0}$:

$$\langle K \rangle = \frac{1}{T} \int_0^T K(t) dt, \quad \langle U \rangle = \frac{1}{T} \int_0^T U(t) dt.$$

We use the facts that over any full period,

$$\langle \sin^2(\omega_0 t + \phi) \rangle = \frac{1}{2}, \quad \langle \cos^2(\omega_0 t + \phi) \rangle = \frac{1}{2}.$$

Therefore,

$$\begin{aligned}\langle K \rangle &= \frac{1}{2}mA^2\omega_0^2 \cdot \frac{1}{2} = \frac{1}{4}mA^2\omega_0^2, \\ \langle U \rangle &= \frac{1}{2}mA^2\omega_0^2 \cdot \frac{1}{2} = \frac{1}{4}mA^2\omega_0^2.\end{aligned}$$

But $E = \frac{1}{2}m\omega_0^2A^2$, so

$$\langle K \rangle = \frac{E}{2}, \quad \langle U \rangle = \frac{E}{2}.$$

Conclusion: over one complete cycle, the oscillator spends half its energy in kinetic form (on average) and half in potential form. The averages are equal and each is $E/2$.

3. Pendulum with a horizontally accelerating support.

We have a pendulum of length L and bob mass m . Initially it hangs straight down. Then the support point begins accelerating horizontally with constant acceleration a (say to the right). After transients, the pendulum reaches a new equilibrium at some constant angle θ from the vertical, and then small oscillations about that equilibrium are possible.

We will work in the non-inertial frame accelerating with the support. In that frame, there is an effective “pseudo-force” of magnitude ma acting on the bob opposite the acceleration of the support (i.e. to the left if the support accelerates to the right). Gravity mg acts downward.

(a) *New equilibrium angle θ .*

In the accelerating frame, equilibrium means the tension $\vec{T}$ in the string balances the vector sum of gravity and the horizontal pseudo-force. Let θ be the angle the string makes with the vertical, displaced in the *opposite* direction of the pseudo-force so that the bob is “behind” the accelerating support.

Force balance requires that the string aligns with the resultant of mg downward and ma horizontally. Geometrically,

$$\tan \theta = \frac{\text{horizontal pseudo-force}}{\text{gravity}} = \frac{ma}{mg} = \frac{a}{g}.$$

Thus

$$\theta = \arctan\left(\frac{a}{g}\right).$$

(b) *Period of small oscillations about this equilibrium.*

For small oscillations about the new equilibrium, the pendulum behaves like a simple pendulum of length L in a uniform effective gravitational field $\vec{g}_{\text{eff}}$ equal to the vector sum of the real gravity and the pseudo-gravity from acceleration.

The magnitude of this effective gravitational field is

$$g_{\text{eff}} = \sqrt{g^2 + a^2}.$$

The restoring torque for small angular displacements $\delta\theta$ about equilibrium is proportional to $mg_{\text{eff}}L\delta\theta$, so the small-angle equation of motion is that of a simple pendulum with g replaced by g_{eff} .

Therefore, the small-oscillation angular frequency is

$$\omega_{\text{small}} = \sqrt{\frac{g_{\text{eff}}}{L}} = \sqrt{\frac{\sqrt{g^2 + a^2}}{L}}.$$

The corresponding period is

$$T = \frac{2\pi}{\omega_{\text{small}}} = 2\pi \sqrt{\frac{L}{g_{\text{eff}}}} = 2\pi \sqrt{\frac{L}{\sqrt{g^2 + a^2}}}.$$

Check limiting cases:

- If $a = 0$, then $g_{\text{eff}} = g$ and $T = 2\pi\sqrt{L/g}$, which is the usual pendulum period.
- If a is large, the effective gravity increases, so the period decreases.

4. Damped harmonic oscillator amplitude decay.

The position of a damped oscillator is

$$x(t) = Ae^{-\beta t} \cos(\omega_1 t + \phi),$$

where A is the initial amplitude factor, $\beta > 0$ is the damping coefficient (with units of 1/time), and ω_1 is the damped angular frequency.

The amplitude envelope of the oscillation is

$$A_{\text{env}}(t) = Ae^{-\beta t}.$$

We are asked: how long until the amplitude decreases to half its initial value?

We want $t = t_{1/2}$ such that

$$A_{\text{env}}(t_{1/2}) = \frac{A}{2}.$$

That is,

$$Ae^{-\beta t_{1/2}} = \frac{A}{2} \quad \implies \quad e^{-\beta t_{1/2}} = \frac{1}{2}.$$

Take the natural logarithm:

$$-\beta t_{1/2} = \ln\left(\frac{1}{2}\right) = -\ln 2,$$

so

$$t_{1/2} = \frac{\ln 2}{\beta}.$$

This $t_{1/2}$ is the amplitude “half-life” for the damped oscillator. Note that it depends only on β , not on ω_1 or A .